Chapter 2 Electrostatics

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2.1 Fundamentals of Electrostatic Fields

In this section, we first introduce Coulomb's law as a fundamental postulate of electrostatics. We then derive the expressions resulting from applying the divergence and curl operators to the electric field, leading to Gauss's law and the formulation of the electric scalar potential.

2.1.1 Coulomb's Law

Coulomb's law is a fundamental principle in electrostatics that describes the force between two point charges. Suppose the first (source) charge q_s is located at $\vec{r}' = x'\hat{x} + y'\hat{y} + z'\hat{z}$ and the second (test) charge q_t is located at $\vec{r} = x\hat{x} + y\hat{y} + z\hat{z}$. We define the distance vector as

$$\vec{R} = \vec{r} - \vec{r}' = R\hat{R} \tag{2.1.1}$$

where $R = \sqrt{(x-x')^2 + (y-y')^2 + (z-z')^2}$ and $\hat{R}$ is the unit vector pointing from q_s to q_t. Coulomb's law states that the electrostatic force $\vec{F}_e$

exerted by q_s to q_t in vacuum is proportional to the product of the charges and inversely proportional to the square of the distance between them:

$$\vec{F_e} = \frac{q_s q_t}{4\pi\epsilon_0 R^2}\hat{R} \tag{2.1.2}$$

where

$$\epsilon_0 \cong 8.854 \times 10^{-12} \cong \frac{1}{36\pi} \times 10^{-9} \quad (\mathrm{F/m}) \tag{2.1.3}$$

is the vacuum permittivity. Accordingly, the electric field is defined as the force experienced by the second charge (test charge) per unit charge:

$$\vec{E} = \frac{\vec{F_e}}{q_t} = \frac{q_s}{4\pi\epsilon_0 R^2}\hat{R} \tag{2.1.4}$$

The definition in (2.1.4) can be generalized to cases involving a distribution of charges. For discrete set of charges, suppose we have a total of N discrete charges presented in space, then by linearilty the electric field can be expressed as

$$\vec{E} = \sum_{n=1}^{N} \frac{q_n}{4\pi\epsilon_0 R_n^2}\hat{R}_n \tag{2.1.5}$$

where R_n is the distance bewteen the test charge and the discrete charges. For continuous set of charges, we have

$$\vec{E} = \int_{V'} \frac{\rho(\vec{r}')}{4\pi\epsilon_0 R^2}\hat{R}dv' \tag{2.1.6}$$

The relation bewteen the total source charge Q and $\rho(\vec{r}')$[1] is given by

$$Q = \int_{V'} \rho(\vec{r}')dv' \tag{2.1.7}$$

[1] Note that the same symbol ρ is used for both the radial component in cylindrical coordinates and the volume charge density, following common conventions. The context will clarify the meaning when necessary.

2.1.2 Gauss' Law

Gauss' law is a fundamental principle in electrostatics that relates the electric flux through a closed surface to the total electric charge enclosed within that surface. To derive Gauss' law, let us take the divergence of the electric field (2.1.6)

$$\vec{\nabla} \cdot \vec{E} = \int_{V'} \frac{\rho(\vec{r}')}{4\pi\epsilon_0} \vec{\nabla} \cdot \left(\frac{\hat{R}}{R^2}\right) dv' \tag{2.1.8}$$

The divergence operator does not act on the source charge distribution $\vec{r}'$, since ρ depends on $\vec{r}'$, whereas $\vec{\nabla} \cdot$ operates with respect to $\vec{r}$. Now, let us evaluate $\vec{\nabla} \cdot \left(\hat{R}/R^2\right)$. We set $\vec{r}'$ to be the new origin of the spherical coordinate (R, θ, ϕ), and consider first the case when $\vec{r} \neq \vec{r}'$,

$$\vec{\nabla} \cdot \left(\frac{\hat{R}}{R^2}\right) = \frac{1}{R^2} \partial_R \left(R^2 \cdot \frac{1}{R^2}\right) = 0, \quad \text{for } R \neq 0 \tag{2.1.9}$$

When $\vec{r} = \vec{r}'$, we have a singularity at the new origin, and special care must be taken to avoid this. We first apply the divergence theorem (1.3.23)

$$\int_V \vec{\nabla} \cdot \left(\frac{\hat{R}}{R^2}\right) dv = \oint_S \frac{\hat{R}}{R^2} \cdot d\vec{s} \tag{2.1.10}$$

We choose the volume V to be an infinitesimal small sphere with its origin at $\vec{r}'$ and its radius equal to dR. Then, the surface integral in (2.1.10) becomes

$$\int_0^{2\pi} \int_0^{\pi} \frac{\hat{R}}{dR^2} \cdot \hat{R} dR^2 \sin\theta \, d\theta d\phi = 4\pi \tag{2.1.11}$$

Let $dR \to 0$, we get

$$\int_V \vec{\nabla} \cdot \left(\frac{\hat{R}}{R^2}\right) dv = 4\pi, \quad \text{for } R \to 0 \tag{2.1.12}$$

Thus, we can express the divergence as

$$\vec{\nabla} \cdot \left(\frac{\hat{R}}{R^2}\right) = 4\pi\delta(\vec{R}) \tag{2.1.13}$$

where $\delta(\vec{R})$ is the delta function with the following property

$$\int_V \delta(\vec{R})dv = \begin{cases} 1, & \text{if } V \text{ encloses } \vec{R} = 0 \\ 0, & \text{otherwise} \end{cases} \tag{2.1.14}$$

Plug (2.1.13) into (2.1.8), we get the differential form of Gauss' law:

$$\vec{\nabla} \cdot \vec{E} = \int_{V'} \frac{\rho(\vec{r}')}{4\pi\epsilon_0} 4\pi\delta(\vec{r} - \vec{r}')dv' = \frac{\rho}{\epsilon_0} \tag{2.1.15}$$

Applying the divergence theorem and (2.1.17), we get the integral form of Gauss' law:

$$\int_V \vec{\nabla} \cdot \vec{E} dv = \oint_S \vec{E} \cdot d\vec{s} = \int_V \frac{\rho}{\epsilon_0} dv = \frac{Q}{\epsilon_0} \tag{2.1.16}$$

Mathematically, Gauss' law states that the surface integral of the normal component of the electric field over a closed surface is equal to the total enclosed charge divided by the permittivity of free space. Since Gauss' law involves the inner product between the electric field and the surface normal, analytical solutions are limited to problems with specific symmetrical geometries. Let us consider three fundamental examples utilizing the integral form of Gauss' law.

□ **<u>Example 2-2</u>** <u>(Cheng 3-5)</u>

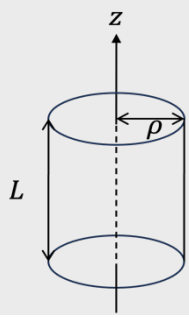
Determine the electric field intensity produced by an infinitely long, straight line charge with uniform linear charge density ρ_l in air.

Solution

Choose a cylindrical coordinate system with the line charge aligned along the z-axis. By symmetry, we can show that the electric field only has the radial component $\vec{E} = E_\rho(\rho)\hat{\rho}$. We choose the Gaussian surface as depicted. Then,

$$\oint_S \vec{E} \cdot d\vec{s} = E_\rho 2\pi\rho L = \int_V \frac{\rho}{\epsilon_0} dv = \frac{\rho_l L}{\epsilon_0}$$

$$\vec{E} = E_\rho \hat{\rho} = \frac{\rho_l}{2\pi\epsilon_0\rho}\hat{\rho}$$



□ **Example 2-3** (Cheng 3-6)

Determine the electric field intensity produced by an infinite plane of charge with uniform surface charge density ρ_s.

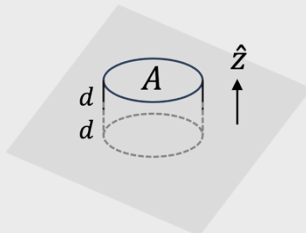
Solution

We place the surface charge on the *xy*-plane. By symmetry, the electric field is directed only along the *z*-axis $\vec{E} = E_z(z)\hat{z}$. Also, the symmetry of the problem also requires $\vec{E}(-z) = -E_z(z)\hat{z}$. We choose the Gaussian surface as depicted. Then,

$$\oint_S \vec{E} \cdot d\vec{s} = E_z(d)\hat{z} \cdot A\hat{z} + E_z(-d)\hat{z} \cdot A(-\hat{z})$$

$$= 2AE_z(d) = \int_V \frac{\rho}{\epsilon_0} dv = \frac{\rho_s A}{\epsilon_0}$$

$$E_z(d) = E_z = \frac{\rho_s}{2\epsilon_0}$$



Thus,

$$\vec{E} = E_z\hat{z} = \begin{cases} \dfrac{\rho_s}{2\epsilon_0}\hat{z}, & z > 0 \\ -\dfrac{\rho_s}{2\epsilon_0}\hat{z}, & z < 0 \end{cases}$$

□ **Example 2-4** (Cheng 3-7)

Determine the electric field produced by a spherical charge distribution with a volume charge density $\rho = \rho_0$ for $0 < r < a$, and $\rho = 0$ for $r > a$.

Solution

It is natural to use the spherical coordinate system for this example, placing the center of the spherical charge distribution at the origin. The Gaussian surface is selected as a sphere with radius equal to r. By symmetry, it is clear that $\vec{E} = E_r(r)\hat{r}$.

For $0 < r < a$

$$\oint_S \vec{E} \cdot d\vec{s} = E_r(r)\pi r^2 = \int_V \frac{\rho}{\epsilon_0} dv = \frac{4}{3\epsilon_0}\pi r^3 \rho_0$$

For $r > a$

$$\oint_S \vec{E} \cdot d\vec{s} = E_r(r)\pi r^2 = \int_V \frac{\rho}{\epsilon_0} dv = \frac{4}{3\epsilon_0}\pi a^3 \rho_0$$

Thus,

$$\vec{E} = E_r(r)\hat{r} = \begin{cases} \frac{4}{3\epsilon_0} r\rho_0 \hat{r}, & 0 < r < a \\ \frac{4a^3}{3\epsilon_0 r^2}\rho_0 \hat{r}, & r > a \end{cases}$$

2.1.3 Electric Scalar Potential

The electric scalar potential, is a fundamental concept in electrostatics that provides a scalar description of the electric field. Let us first take the curl of the electric field (2.1.6)

$$\vec{\nabla} \times \vec{E} = \int_{V'} \frac{\rho(\vec{r}')}{4\pi\epsilon_0} \vec{\nabla} \times \left(\frac{\hat{R}}{R^2}\right) dv' = 0 \tag{2.1.17}$$

or in integral from

$$\int_s \vec{\nabla} \times \vec{E} \cdot d\vec{s} = \oint_C \vec{E} \cdot d\vec{l} = 0 \tag{2.1.18}$$

To show that (2.1.17) is identically zero, one can apply Example 1.6(b). Thus, since the electrostatic field is irrotational, from the null identity (1.3.28), we can express the electric field as

$$\vec{E} = -\vec{\nabla}\mathcal{V} \tag{2.1.19}$$

We call the scalar function $\mathcal{V}$ as the electric scalar potential. The minus sign in (2.1.19) arises from the convention that the electric field points from regions of higher to lower potential. From the gradient property (1.3.3),

$$dV = \vec{\nabla}\mathcal{V} \cdot d\vec{l} = -\vec{E} \cdot d\vec{l} \tag{2.1.20}$$

Integrating (2.1.20) from point A to point B, we get

$$\mathcal{V}_B - \mathcal{V}_A = \Delta\mathcal{V} = -\int_A^B \vec{E} \cdot d\vec{l} = \int_B^A \vec{E} \cdot d\vec{l} \tag{2.1.21}$$

$\Delta\mathcal{V}$ is the potential difference between B and A. It is meaningless to speak of absolute potential. Typically, zero potential is defined at infinity, but if another reference is used, it must be explicitly stated. (2.1.21) also shows that the integral $\int_A^B \vec{E} \cdot d\vec{l}$ is path-independent in electrostatics.

To derive the general expression for electric scalar potential, let us consider the following vector result first. Since

$$\begin{aligned}\vec{\nabla}\left(\frac{1}{R}\right) &= \vec{\nabla}\left[\frac{1}{\sqrt{(x-x')^2+(y-y')^2+(z-z')^2}}\right] \\ &= -\frac{(x-x')\hat{x}+(y-y')\hat{y}+(z-z')\hat{z}}{[(x-x')^2+(y-y')^2+(z-z')^2]^{\frac{3}{2}}} \\ &= -\frac{\vec{R}}{R^3} = -\frac{\hat{R}}{R^2}\end{aligned} \tag{2.1.22}$$

Thus,

$$\vec{E} = -\int_{V'} \frac{\rho(\vec{r}')}{4\pi\epsilon_0}\left(-\frac{\hat{R}}{R^2}\right) dv' = -\vec{\nabla}\int_{V'} \frac{\rho(\vec{r}')}{4\pi\epsilon_0 R} dv' \tag{2.1.23}$$

By comparing (2.1.19) and (2.1.23), we can get the expression of the scalar electric potential:

$$\mathcal{V} = \int_{V'} \frac{\rho(\vec{r}')}{4\pi\epsilon_0 R} dv' \tag{2.1.24}$$

It is evident that evaluating the scalar potential by (2.1.24) is much simpler than direct calculation of the electric field by (2.1.6) since it involves no vectors. Therefore, it is often advantageous to first calculate the electric scalar potential and then use (2.1.19) to obtain the electric field.

□ **<u>Example 2-5</u>** <u>(Cheng 3-10)</u>

Derive an expression for the electric field intensity along the axis of a uniformly charged line segment of length L, with a constant line charge density ρ_l.

Solution

For $|z| > L/2$

$$\mathcal{V}(z) = \int_{-L/2}^{L/2} \frac{\rho_l}{4\pi\epsilon_0 (z - z')} dz' = \frac{\rho_l}{4\pi\epsilon_0} \int_{-L/2}^{L/2} \frac{1}{(z - z')} dz'$$

$$= \frac{\rho_l}{4\pi\epsilon_0} \ln\left(\frac{z + L/2}{z - L/2}\right)$$

Then,

$$\vec{E}(z) = -\vec{\nabla}\mathcal{V} = -\frac{d\mathcal{V}}{dz}\hat{z} = \frac{\rho_l}{4\pi\epsilon_0}\left(\frac{1}{z - L/2} - \frac{1}{z + L/2}\right)\hat{z}$$

$$= \frac{\rho_l}{4\pi\epsilon_0} \frac{L}{z^2 - (L/2)^2}\hat{z}$$

□ **<u>Example 2-6</u>** <u>(Cheng 3-9)</u>

Derive an expression for the electric field intensity along the axis of a circular disk of radius a with a uniform surface charge density ρ_s.

Solution

$$\mathcal{V}(z) = \int_0^{2\pi} \int_0^a \frac{\rho_s}{4\pi\epsilon_0 \sqrt{z^2 + r'^2}} r' dr' d\phi' = \frac{\rho_s}{2\epsilon_0} \int_0^a \frac{r'}{\sqrt{z^2 + r'^2}} dr'$$

$$= \frac{\rho_s}{2\epsilon_0} \int_0^{a^2} \frac{du}{2\sqrt{z^2 + u}} = \frac{\rho_s}{2\epsilon_0} \sqrt{z^2 + u}\Big|_0^{a^2}$$

$$= \frac{\rho_s}{2\epsilon_0}\left(\sqrt{z^2 + a^2} - |z|\right)$$

Then,

$$\vec{E}(z) = -\vec{\nabla}\mathcal{V} = -\frac{d\mathcal{V}}{dz}\hat{z} = \begin{cases} \dfrac{\rho_s}{2\epsilon_0}\left(1 - \dfrac{z}{\sqrt{z^2 + a^2}}\right)\hat{z}, & z > 0 \\ -\dfrac{\rho_s}{2\epsilon_0}\left(1 + \dfrac{z}{\sqrt{z^2 + a^2}}\right)\hat{z}, & z < 0 \end{cases}$$

2.2 Electric Dipole

Electric dipoles are fundamental in understanding the behavior of molecules in electric fields, dielectric polarization. The electric field generated by a dipole falls off more rapidly with distance than that of a single point charge and has a distinctive directional dependence. Consider a simple electric dipole consisting of a pair of equal and opposite charges, $+q$ and $-q$, located at $(0,0,d/2)$ and $(0,0,-d/2)$ in Cartesian coordinates. We will determine the resulting electric field at the position $\vec{r}$ when $r \gg d$. Let $\vec{d} = (0,0,d)$, we can express electric potential as

$$\mathcal{V} = \frac{q}{4\pi\epsilon_0}\left(\frac{1}{|\vec{r}-\vec{d}/2|} - \frac{1}{|\vec{r}+\vec{d}/2|}\right) \tag{2.2.1}$$

Consider

$$|\vec{r}-\vec{d}/2|^{-1} = \left[(\vec{r}-\vec{d}/2)\cdot(\vec{r}-\vec{d}/2)\right]^{-\frac{1}{2}} = \left(r^2 - \vec{r}\cdot\vec{d} - d^2/4\right)^{-\frac{1}{2}}$$

$$\cong r^{-1}\left(1 - \frac{\vec{r}\cdot\vec{d}}{r^2}\right)^{-\frac{1}{2}} \cong r^{-1}\left(1 + \frac{1}{2}\frac{\vec{r}\cdot\vec{d}}{r^2}\right)$$

Similarly,

$$|\vec{r}+\vec{d}/2|^{-1} \cong r^{-1}\left(1 - \frac{1}{2}\frac{\vec{r}\cdot\vec{d}}{r^2}\right)$$

Define the electric dipole moment as

$$\vec{p} = q\vec{d} = p\hat{z} = p\left(\cos\theta\,\hat{r} - \sin\theta\,\hat{\theta}\right) \tag{2.2.2}$$

Thus, the electric potential can be approximate as

$$\mathcal{V} \cong \frac{q}{4\pi\epsilon_0 r}\left(\frac{\vec{r}\cdot\vec{d}}{r^2}\right) = \frac{1}{4\pi\epsilon_0 r^2}\vec{p}\cdot\hat{r} = \frac{p\cos\theta}{4\pi\epsilon_0 r^2} \tag{2.2.3}$$

The electric field can then be computed by

$$\begin{aligned}\vec{E} = -\vec{\nabla}\mathcal{V} &\cong -\left(\hat{r}\partial_r V + \hat{\theta}\frac{1}{r}\partial_\theta V\right)\\ &= \frac{p}{4\pi\epsilon_0 r^3}\left[2\cos\theta\,\hat{r} + \sin\theta\,\hat{\theta}\right]\\ &= \frac{1}{4\pi\epsilon_0 r^3}\left[3(\hat{r}\cdot\vec{p})\hat{r} - \vec{p}\right]\end{aligned} \tag{2.2.4}$$

2.3 Electrostatics Concerning Materials

In the previous section, we examined electrostatics in free space. In this section, we consider electrostatics in the presence of matter.

2.3.1. Dielectrics

Good dielectrics have tightly bound charges that prevent current flow. Under an external electric field, Coulomb forces displace nuclei and electron clouds, creating induced dipoles. Consider a differential volume dv' containing a dipole moment per unit volume (known as polarization vector) $\vec{P}(\vec{r}')$. Then, the total dipole moment inside this differential volume is $\vec{P}(\vec{r}')dv'$. According to (2.2.3), the differential electric potential at $\vec{r}$ is

$$d\mathcal{V} = \frac{\vec{P}(\vec{r}') \cdot \hat{R}}{4\pi\epsilon_0 R^2} dv' \tag{2.3.1}$$

where R is defined in (2.1.1). Integrating (2.3.1), and from (2.1.22)[2], we get

$$\mathcal{V} = \int_{V'} \frac{\vec{P}(\vec{r}') \cdot \hat{R}}{4\pi\epsilon_0 R^2} dv' = \frac{1}{4\pi\epsilon_0} \int_{V'} \vec{P}(\vec{r}') \cdot \vec{\nabla}' \left(\frac{1}{R}\right) dv' \tag{2.3.2}$$

Also, by noting the identity

$$\vec{P}(\vec{r}') \cdot \vec{\nabla}' \left(\frac{1}{R}\right) = \vec{\nabla}' \cdot \left[\frac{\vec{P}(\vec{r}')}{R}\right] - \frac{\vec{\nabla}' \cdot \vec{P}(\vec{r}')}{R} \tag{2.3.3}$$

we can rewrite (2.3.2) as

$$\begin{aligned} \mathcal{V} &= \frac{1}{4\pi\epsilon_0} \left\{ \int_{V'} \vec{\nabla}' \cdot \left[\frac{\vec{P}(\vec{r}')}{R}\right] dv' + \int_{V'} \frac{-\vec{\nabla}' \cdot \vec{P}(\vec{r}')}{R} dv' \right\} \\ &= \frac{1}{4\pi\epsilon_0} \left[\oint_{S'} \frac{\vec{P}(\vec{r}') \cdot \hat{n}}{R}' ds' + \int_{V'} \frac{-\vec{\nabla}' \cdot \vec{P}(\vec{r}')}{R} dv' \right] \\ &= \frac{1}{4\pi\epsilon_0} \left[\oint_{S'} \frac{\rho_{ps}(\vec{r}')}{R} ds' + \int_{V'} \frac{\rho_p(\vec{r}')}{R} dv' \right] \end{aligned} \tag{2.3.4}$$

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[2] One can show that $\vec{\nabla}\left(\frac{1}{R}\right) = -\vec{\nabla}'\left(\frac{1}{R}\right)$, where $\vec{\nabla}'$ denotes operation with respect to $\vec{r}'$.

where $\hat{n}'$ is the outward normal vector on S' and ρ_p / ρ_{ps} are the polarization charge density / polarization surface charge density with

$$\rho_p(\vec{r}') = -\vec{\nabla}' \cdot \vec{P}(\vec{r}') \tag{2.3.5}$$

and

$$\rho_{ps}(\vec{r}') = \vec{P}(\vec{r}') \cdot \hat{n}' \tag{2.3.6}$$

Bound charges associated with ρ_p and ρ_{ps} are not free to move, unlike free charges.

Consider Gauss' law when there are both bound charge density ρ_p and free charge density ρ:

$$\vec{\nabla} \cdot \vec{E} = \frac{(\rho + \rho_p)}{\epsilon_0} \tag{2.3.7}$$

Substituting (2.3.5) in (2.3.7) and rearrange the terms, we get

$$\vec{\nabla} \cdot \left(\epsilon_0 \vec{E} + \vec{P}\right) = \rho \tag{2.3.8}$$

Now, let us introduce the electric flux density

$$\vec{D} = \epsilon_0 \vec{E} + \vec{P} \tag{2.3.9}$$

then

$$\vec{\nabla} \cdot \vec{D} = \rho \tag{2.3.10}$$

Or in integral form

$$\oint_S \vec{D} \cdot d\vec{s} = Q \tag{2.3.11}$$

where Q denotes the total free charges enclosed by S. Note that (2.3.10-11) involves only the free charges; equivalently, the definition of $\vec{D}$ already contains the effect of polarization.

In most cases (linear[3] and isotropic[4] materials), we have

$$\vec{P} = \chi_e \epsilon_0 \vec{E} \tag{2.3.12}$$

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[3] $\vec{P}$ does not contain higher order terms of $\vec{E}$, i.e., $|\vec{E}|^2$ or $|\vec{E}|^3$, etc.

[4] The direction of $\vec{P}$ aligns with the direction of $\vec{E}$.

where χ_e is a dimensionless quantity called electric susceptibility, and the relation between $\vec{D}$ and $\vec{E}$ is given by

$$\vec{D} = \epsilon_0(1+\chi_e)\vec{E} = \epsilon_0\epsilon_r\vec{E} = \epsilon\vec{E} \tag{2.3.13}$$

ϵ_r is called the relative permittivity and ϵ is called the absolute permittivity. If ϵ is independent of the position, we say that the medium is homogenous, and a linear, isotropic, and homogenous medium is often called a simple medium.

□ **<u>Example 2-7</u>** <u>(Cheng 3-12)</u>

A positive point charge q is placed at the center of a spherical dielectric shell with inner radius r_i and outer radius r_o, where the shell has a dielectric constant ϵ_r. Determine $\vec{E}, \mathcal{V}, \vec{D}, \vec{P}$ as functions of the radial distance r.

Solution

Use spherical coordinate for analysis, and note that by symmetry, $\vec{E} = E_r(r)\hat{r}$, etc. Consider the following three cases, and apply Gauss' law in integral form

(a) $r_0 < r$

$$D_r = \frac{q}{4\pi r^2}, \qquad E_r = \frac{D_r}{\epsilon_0} = \frac{q}{4\pi\epsilon_0 r^2}$$

$$P_r = D_r - \epsilon_0 E_r = 0$$

$$\mathcal{V} = \frac{q}{4\pi\epsilon_0 r}$$

(b) $r_i < r < r_o$

$$D_r = \frac{q}{4\pi r^2}, \qquad E_r = \frac{D_r}{\epsilon} = \frac{q}{4\pi\epsilon_0\epsilon_r r^2}$$

$$P_r = D_r - \epsilon_0 E_r = \frac{q}{4\pi r^2}\left(1 - \frac{1}{\epsilon_r}\right)$$

$$\mathcal{V} = -\int_{\infty}^{r_o} E_r dr - \int_{r_o}^{r} E_r dr = \frac{q}{4\pi\epsilon_0 r_o} - \frac{q}{4\pi\epsilon_0\epsilon_r}\int_{r_o}^{r}\frac{1}{r^2}dr$$

$$= \frac{q}{4\pi\epsilon_0}\left[\frac{1}{\epsilon_r r} + \frac{1}{r_o}\left(1 - \frac{1}{\epsilon_r}\right)\right]$$

(c) $r < r_i$

$$D_r = \frac{q}{4\pi r^2}, \qquad E_r = \frac{D_r}{\epsilon_0} = \frac{q}{4\pi\epsilon_0 r^2}$$

$$P_r = D_r - \epsilon_0 E_r = 0$$

$$\mathcal{V} = -\int_{\infty}^{r_i} E_r dr - \int_{r_i}^{r} E_r dr = \frac{q}{4\pi\epsilon_0}\left[\frac{1}{r_i}\left(\frac{1}{\epsilon_r} - 1\right) + \frac{1}{r_o}\left(1 - \frac{1}{\epsilon_r}\right) + \frac{1}{r}\right]$$

2.3.2. Conductors

A conductor is a material in which electric charges can move freely. When an external electric field is applied to a conductor, free charges within it quickly redistribute themselves in such a way that the internal electric field is canceled. As a result, the electric field inside a conductor in electrostatic equilibrium is zero. This redistribution of charge also causes all excess charges to reside on the surface of the conductor. Thus, under static conditions, we have the no free charges inside a conductor (aside from surface charge density ρ_s)

$$\rho = 0 \tag{2.3.14}$$

And by Gauss's law, the internal electric field is also zero

$$\vec{E} = 0 \tag{2.3.15}$$

Furthermore, since we have zero fields inside the conductor, the conductor becomes an equipotential body, meaning the electric potential remains constant throughout its volume and on its surface.

□ **<u>Example 2-8</u>** <u>(Cheng 3-11)</u>

Consider the same configuration in Example 2-7, except that the dielectric material is replaced by a conductor. Find $\vec{E}$ and $\mathcal{V}$.

Solution

(a) $r_0 < r$

$$E_r = \frac{q}{4\pi\epsilon_0 r^2}$$

$$\mathcal{V} = \frac{q}{4\pi\epsilon_0 r}$$

(b) $r_i < r < r_o$

$$E_r = 0$$

$$\mathcal{V} = -\int_{\infty}^{r_o} E_r dr - \int_{r_o}^{r} E_r dr = \frac{q}{4\pi\epsilon_0 r_o}$$

(c) $r < r_i$

$$E_r = \frac{q}{4\pi\epsilon_0 r^2}$$

$$\mathcal{V} = -\int_{\infty}^{r_i} E_r dr - \int_{r_i}^{r} E_r dr = \frac{q}{4\pi\epsilon_0}\left[\frac{1}{r_o} - \frac{1}{r_i} + \frac{1}{r}\right]$$

2.4 Boundary Conditions for Electrostatic Fields

Consider the boundary separating two distinct media, say, medium 1 and 2. We are interested in the behavior of the fields across the boundary. Let us first construct a small loop as shown in Fig. 1.1.

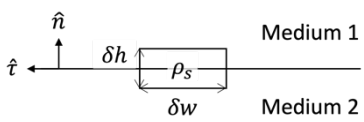


Fig. 1.1
Boundary between 2 media

Integrating the electric field along the loop and let $\delta h \to 0$, we get

$$\oint_C \vec{E} \cdot d\vec{l} = \vec{E}_1 \cdot \delta w \hat{\tau} + \vec{E}_1 \cdot \delta w(-\hat{\tau}) = 0 \qquad (2.4.1)$$

The last equality holds because of (2.1.18). Therefore

$$\vec{E}_1 \cdot \hat{\tau} = \vec{E}_2 \cdot \hat{\tau} \tag{2.4.2}$$

which states that the tangential electric field across the boundary is continuous since the selection of $\hat{\tau}$ can be arbitrary as long as it is perpendicular to boundary surface normal $\hat{n}$. To be more general, we can express (2.4.2) as

$$\hat{n} \times \vec{E}_1 = \hat{n} \times \vec{E}_2 \tag{2.4.3}$$

The result of $\hat{n} \times (\cdot)$ is perpendicular to $\hat{n}$, so only the tangential components remain.

Now let us extend the loop in the $\hat{n} \times \hat{\tau}$ direction with a depth equal to δs to form a small volume. Then, by applying (2.3.11) and also let $\delta h \to 0$

$$\oint_S \vec{D} \cdot d\vec{s} = \vec{D}_1 \cdot \delta w \delta s\, \hat{n} + \vec{D}_2 \cdot \delta w \delta s(-\hat{n}) = Q = \rho_s \delta w \delta s \tag{2.4.4}$$

By some simplification, we get

$$\hat{n} \cdot \left(\vec{D}_1 - \vec{D}_2\right) = \rho_s \tag{2.4.5}$$

which states that the normal component of $\vec{D}$ is discontinuous across the boundary when (free) surface charge density exists. In general, ρ_s exists only when at least one of the adjoining media is a conductor.

Without loss of generality, assume medium 2 is a conductor, and medium 1 is a general dielectric with its permeability equal to ϵ. Under static conditions, the electric field inside a conductor is zero, as stated in (2.3.15). Therefore, from (2.4.3) and (2.4.5), we have:

$$\hat{n} \times \vec{E}_1 = 0 \tag{2.4.6}$$

and

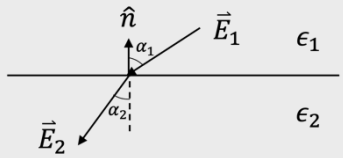
$$\hat{n} \cdot \vec{D}_1 = \hat{n} \cdot \epsilon \vec{E}_1 = \rho_s \tag{2.4.7}$$

Hence, the tangential component of the electric field at the surface of a conductor is zero, and the normal component of the electric displacement field equals the surface charge density.

□ **Example 2-9** (Cheng 3-15)

Two dielectric media with permittivities ϵ_1 and ϵ_2 are separated by a charge-free boundary, as shown in the following figure. The electric field intensity at point P in medium 1 has magnitude $\vec{E}_1$ and forms an angle α_1 with the normal. Determine the magnitude and direction of the electric field intensity at point P in medium 2.

Solution



From (2.4.2) and (2.4.5) we have

$$E_1 \sin\alpha_1 = E_2 \sin\alpha_2$$

$$\epsilon_1 E_1 \cos\alpha_1 = \epsilon_2 E_2 \cos\alpha_2$$

Dividing the two:

$$\frac{\tan\alpha_2}{\tan\alpha_1} = \frac{\epsilon_2}{\epsilon_1} \rightarrow \alpha_2 = \operatorname{atan}\left(\frac{\epsilon_2}{\epsilon_1}\tan\alpha_1\right)$$

And the magnitude of $\vec{E}_2$ is

$$E_2 = \sqrt{(E_2 \sin\alpha_2)^2 + (E_2 \cos\alpha_2)^2}$$

$$= \sqrt{(E_1 \sin\alpha_1)^2 + \left(\frac{\epsilon_1}{\epsilon_2} E_1 \cos\alpha_1\right)^2}$$

$$= E_1 \sqrt{(\sin\alpha_1)^2 + \left(\frac{\epsilon_1}{\epsilon_2}\cos\alpha_1\right)^2}$$

2.5 Capacitance

A capacitor is a fundamental device in electrostatics that stores electric charge and energy. It consists of two (or more) conductors separated by an insulating medium. When a potential difference $\Delta\mathcal{V}$ is applied across the conductors, equal and opposite charges $\pm Q$ accumulate on them. The amount of charge stored is directly proportional to the applied voltage, expressed as:

$$Q = C\Delta\mathcal{V} \tag{2.5.1}$$

where C is the capacitance, a constant that depends on the geometry of the conductors and the permittivity of the insulating material between them.

To determine the capacitance of a given two-conductor configuration, the procedure is as follows. First, choose a coordinate system that best fits the symmetry of the geometry. Then, assume charges $+Q$ and $-Q$ on the two conductors. Using this charge distribution, determine the electric field $\vec{E}$ by applying (2.4.7), Gauss's law, or other relevant relations. Next, compute the potential difference $\Delta\mathcal{V}$ by integrating the electric field along a path from the negatively charged conductor to the positively charged one, using (2.1.21). Finally, obtain the capacitance by taking the ratio $C = Q/\Delta\mathcal{V}$. Let us illustrate this by the following examples:

□ **<u>Example 2-10</u>** <u>(Cheng 3-17)</u>

A parallel-plate capacitor consists of two conducting plates, each with area A, separated by a uniform distance d. The space between the plates is filled with a dielectric of constant permittivity ϵ. Determine its capacitance.

Solution

We put $\pm Q$ on the top/bottom plate. By (2.4.7), the magnitude of the electric field is

$$E = \frac{\rho_s}{\epsilon} = \frac{Q}{A\epsilon}$$

with its direction pointing from top to bottom. The voltage difference is

$$\Delta\mathcal{V} = -\int_d^0 E dl = \int_0^d E dl = \frac{Qd}{A\epsilon}$$

Then,

$$C = \frac{Q}{\Delta\mathcal{V}} = \frac{A\epsilon}{d}$$

□ **<u>Example 2-11</u>** <u>(Cheng 3-18)</u>

A cylindrical capacitor consists of an inner conductor with radius a and a coaxial outer conductor radius b. The region is filled with a dielectric of permittivity ϵ and the capacitor has length L. Determine its capacitance.

Solution

We put $\pm Q$ on the inner/outer cylinder. By Gauss' law

$$\vec{E} = \frac{Q}{2\pi\epsilon L\rho}\hat{\rho}$$

inside the coaxial, where cylindrical coordinate is used.

$$\Delta\mathcal{V} = \int_a^b \vec{E}\cdot d\rho\hat{\rho} = \frac{Q}{2\pi\epsilon L}\ln\frac{b}{a}$$

$$C = \frac{Q}{\Delta\mathcal{V}} = 2\pi\epsilon L\ln\frac{a}{b}$$

□ **<u>Example 2-12</u>** <u>(Cheng 3-19)</u>

A spherical capacitor consists of an inner conducting sphere of radius a and a concentric outer spherical conductor with radius b. The region is filled with a dielectric of permittivity ϵ. Determine the capacitance.

Solution

We put $\pm Q$ on the inner/outer sphere. By Gauss' law

$$\vec{E} = \frac{Q}{4\pi\epsilon r^2}\hat{r}$$

between the spheres, where spherical coordinate is used.

$$\Delta\mathcal{V} = \int_a^b \vec{E}\cdot dr\hat{r} = \frac{Q}{4\pi\epsilon}\left(\frac{1}{a}-\frac{1}{b}\right)$$

$$C = \frac{Q}{\Delta\mathcal{V}} = 4\pi\epsilon/\left(\frac{1}{a}-\frac{1}{b}\right)$$

2.6 Electrostatic Energy

Electrostatic energy refers to the potential energy stored in a system of charges due to their positions in an electric field. When assembling charges, work must be done against electrostatic forces, and this work is stored as energy.

Consider the case when there is no charge in free space. W first bring a charge q_1 to the position $\vec{r}_1$ from infinity. No work is done, that is, $W_1 = 0$, since no other charges are presented. To slowly bring another charge q_2 from infinity to the position $\vec{r}_2$—neglecting kinetic energy and radiation effects—the required work is:

$$W_2 = \int_{\infty}^{\vec{r}_2} -q_2\vec{E} \cdot d\vec{l} = \frac{q_1 q_2}{4\pi\epsilon_0 R_{21}} \tag{2.6.1}$$

where R_{21} is the distance between $\vec{r}_2$ and $\vec{r}_1$. We further bring a third charge q_3 to the position $\vec{r}_3$, the additional work to be done is:

$$W_3 = \frac{q_1 q_3}{4\pi\epsilon_0 R_{31}} + \frac{q_2 q_3}{4\pi\epsilon_0 R_{32}} \tag{2.6.2}$$

Continue the process up to N charges, we need a total work of

$$W_e = \sum_{n=1}^{N} W_n = \frac{1}{2}\sum_{n=1}^{N} q_n \sum_{m=1}^{N} \frac{q_m}{4\pi\epsilon_0 R_{mn}}(1-\delta_{mn}) \tag{2.6.3}$$

where δ_{mn} is the Kronecker delta with $\delta_{mn} = 1$ when $m = n$ and $\delta_{mn} = 0$ when $m \neq n$. Now let

$$\mathcal{V}_n = \sum_{m=1}^{N} \frac{q_m}{4\pi\epsilon_0 R_{mn}}(1-\delta_{mn}) \tag{2.6.4}$$

where V_n is interpreted as the voltage at $\vec{r}_n$ due to all of the charges except q_n. Thus, (2.6.3) is simplified as

$$W_e = \frac{1}{2}\sum_{n=1}^{N} q_n \mathcal{V}_n \tag{2.6.5}$$

(2.6.5) represents the energy required to assemble a set of charges, or equivalently, the energy stored in the resulting charge configuration.

Generalize (2.6.5) to continuous charge distributions, we get

$$W_e = \frac{1}{2}\int_V \rho \mathcal{V} dv \tag{2.6.6}$$

$\mathcal{V}$ in the integrand represents the potential due to the total continuous charge distribution within volume. While $\mathcal{V}$ is due to a continuous distribution and $\mathcal{V}_n$ in (2.6.4) is due to discrete charges excluding q_n, they are compatible. This is because the contribution of an infinitesimal charge element ρdv at the point under integration is negligible, just as q_n is excluded in $\mathcal{V}_n$.

□ **<u>Example 2-13</u>** <u>(Cheng 3-22, 3-23)</u>

Determine the energy required to assemble a uniformly charged sphere of radius a with volume charge density ρ.

Solution

To find the energy required, consider the process of adding a spherical layer to an existing sphere with radius r. The existing sphere has the potential at r with

$$\mathcal{V}(r) = \frac{Q(r)}{4\pi\epsilon_0 r} = \frac{\rho\frac{4}{3}\pi r^3}{4\pi\epsilon_0 r} = \frac{\rho r^2}{3\epsilon_0}$$

The additional charge added on is

$$dQ(r) = 4\pi r^2 \rho dr$$

Thus, the total work required is

$$W_e = \int dw = \int \mathcal{V} dQ = \frac{4\pi\rho^2}{3\epsilon_0}\int_0^a r^4 dr = \frac{4\pi\rho^2 a^5}{15\epsilon_0}$$

Note: one can also solve this by applying (2.6.6)

$$W_e = \frac{1}{2}\int_V \rho \mathcal{V} dv = \frac{\rho}{2}\int_0^a \mathcal{V}(r) 4\pi r^2 dr$$

with

$$\mathcal{V} = \int_r^\infty E_r dr = (\int_r^a + \int_a^\infty) E_r dr = \frac{\rho}{6\epsilon_0}(3a^2 - r^2)$$

Since charge density is often not directly known, we can express (2.6.6) in terms of field quantities as follows:

$$W_e = \frac{1}{2}\int_V (\vec{\nabla}\cdot\vec{D})\mathcal{V}dv = \frac{1}{2}\int_V \left[\vec{\nabla}\cdot(\mathcal{V}\vec{D}) - \vec{D}\cdot\vec{\nabla}\mathcal{V}\right]dv$$
$$= \frac{1}{2}\oint_S \mathcal{V}\vec{D}\cdot d\vec{s} + \frac{1}{2}\int_V \vec{D}\cdot\vec{E}dv \qquad (2.6.7)$$

Since the volume V can be arbitrary as long as it includes all the charges, we choose V as a sphere with radius $r \to \infty$. Since $\mathcal{V} \propto 1/r$ and $\vec{D} \propto 1/r^2$, but $S \propto r^2$, the surface integral in (2.6.7) vanishes when $r \to \infty$. Thus,

$$W_e = \frac{1}{2}\int_V \vec{D}\cdot\vec{E}dv = \frac{1}{2}\int_V \epsilon E^2 dv = \frac{1}{2}\int_V \frac{D^2}{\epsilon}dv = \frac{1}{2}\int_V w_e dv \qquad (2.6.8)$$

where

$$w_e = \frac{1}{2}\vec{D}\cdot\vec{E} = \frac{1}{2}\epsilon E^2 = \frac{1}{2\epsilon}D^2 \qquad (2.6.9)$$

is the electrostatic energy density.

□ **<u>Example 2-14</u>** <u>(Cheng 3-24)</u>

From Example 2-10, find the energy stored in a parallel-plate capacitor.

Solution

From Example 2-10 we have

$$E = \frac{Q}{A\epsilon} = \frac{\Delta\mathcal{V}}{d}$$

Applying (2.6.8)

$$W_e = \frac{1}{2}\int_V \epsilon E^2 dv = \frac{1}{2}\int_V \epsilon\left(\frac{\Delta\mathcal{V}}{d}\right)^2 dv = \frac{1}{2}\epsilon\left(\frac{\Delta\mathcal{V}}{d}\right)^2 Ad = \frac{1}{2}\left(\frac{\epsilon A}{d}\right)\Delta\mathcal{V}^2$$
$$= \frac{1}{2}C\Delta\mathcal{V}^2 = \frac{1}{2}Q\Delta\mathcal{V} = \frac{Q^2}{2C}$$

Note: also verify the result for Example 2-11 and 2-12.

2.7 Poisson and Laplace Equations

In electrostatics, Poisson's and Laplace's equations describe how electric potential $\mathcal{V}$ behaves in space. From (2.3.10) and (2.3.13)

$$\vec{\nabla} \cdot \left(\epsilon \vec{E}\right) = -\vec{\nabla} \cdot \left(\epsilon \vec{\nabla} \mathcal{V}\right) = \rho \tag{2.7.1}$$

For homogeneous medium, we can then simplify (2.7.1) to

$$\nabla^2 \mathcal{V} = -\frac{\rho}{\epsilon} \tag{2.7.2}$$

where ∇^2 is the Laplacian operator as defined in (1.3.16). (2.7.2), known as Poisson equation, states that the Laplacian of the potential $\mathcal{V}$—that is, the divergence of its gradient—is equal to $-\rho/\epsilon$, where ϵ is the constant permittivity of a simple medium and ρ is the volume charge density, which may vary with position. In a source-free region, the Poisson equation reduces to the Laplace equation:

$$\nabla^2 \mathcal{V} = 0 \tag{2.7.3}$$

The objective of this section is to determine a unique solution for the electric potential $\mathcal{V}$ that satisfies Poisson or Laplace equation within a given region. However, the differential equation alone does not guarantee uniqueness. The uniqueness of the solution is ensured by applying suitable boundary conditions, that is, the solution of $\mathcal{V}$ to the Poisson equation is unique in the volume V if $\mathcal{V}$ is specified on the boundary S.

Let us assume that two solutions $\mathcal{V}_1$ and $\mathcal{V}_2$ exists in the volume V that satisfy the Poisson equation:

$$\nabla^2 \mathcal{V}_1 = -\frac{\rho}{\epsilon} \tag{2.7.4a}$$

$$\nabla^2 \mathcal{V}_2 = -\frac{\rho}{\epsilon} \tag{2.7.4b}$$

By subtracting the two, we found that their difference $\mathcal{V}_d = \mathcal{V}_2 - \mathcal{V}_1$ satisfies the Laplace equation:

$$\nabla^2 \mathcal{V}_d = 0 \tag{2.7.5}$$

Now, consider the vector identity

$$\vec{\nabla} \cdot \left(f\vec{A}\right) = \vec{\nabla} f \cdot \vec{A} + f\vec{\nabla} \cdot \vec{A} \tag{2.7.6}$$

with $f = \mathcal{V}_d$ and $\vec{A} = \vec{\nabla}\mathcal{V}_d$, we get

$$\vec{\nabla} \cdot \left(\mathcal{V}_d \vec{\nabla}\mathcal{V}_d\right) = \left|\vec{\nabla}\mathcal{V}_d\right|^2 + \mathcal{V}_d \, \nabla^2 \mathcal{V}_d \tag{2.7.7}$$

Integrating (2.7.7) over the volume V, and apply (2.7.5) and the divergence theorem:

$$\oint_S \mathcal{V}_d \vec{\nabla}\mathcal{V}_d \cdot d\vec{s} = \oint_S \mathcal{V}_d \partial_n \mathcal{V}_d ds = \int_V \left|\vec{\nabla}\mathcal{V}_d\right|^2 dv \tag{2.7.8}$$

where $\partial_n \mathcal{V}_d = \vec{\nabla}\mathcal{V}_d \cdot d\hat{n}$ is the normal derivative on the surface S. Three common types of boundary conditions arise in electrostatic boundary-value problems:

i. **Dirichlet boundary condition**: $\mathcal{V}$ is specified over the surface S. In this case, the boundary values of the potential are known ($\mathcal{V}_d = \mathcal{V}_2 - \mathcal{V}_1 = 0$ over S).
ii. **Neumann boundary condition**: $\partial_n \mathcal{V}$ is specified over the surface S ($\partial_n \mathcal{V}_d = 0$ over S). This corresponds to prescribing the normal component of the electric field.
iii. **Mixed boundary condition**: $\mathcal{V}$ is specified on a portion of the boundary $S_1 \subset S$, while the normal derivative $\partial_n \mathcal{V}$ is specified on the remaining part $S_2 = S/S_1$.

In all cases, we have $\oint_S \mathcal{V}_d \partial_n \mathcal{V}_d ds = 0$. Thus, (2.7.8) can be simplified as

$$\int_V \left|\vec{\nabla}\mathcal{V}_d\right|^2 dv = 0 \tag{2.7.9}$$

Since $\left|\vec{\nabla}\mathcal{V}_d\right|^2$ is non-negative everywhere in V, we can conclude that $\left|\vec{\nabla}\mathcal{V}_d\right| = 0$ everywhere in V. For Dirichlet boundary condition, we have $\mathcal{V}_d = \mathcal{V}_2 - \mathcal{V}_1 = 0$, or $\mathcal{V}_2 = \mathcal{V}_1$ everywhere in V, that is, the solution in unique. For Neumann boundary condition, we have $\mathcal{V}_2 = \mathcal{V}_1 + C$ everywhere in V, where C is an additive constant. In this case, we can also conclude that the unique solution is $\mathcal{V}_1 + C$.

Now, let us solve the boundary-value problem in Cartesian coordinates by using the method of separation of variables. Here we consider the Laplace equation. By expressing (2.7.3) in Cartesian coordinate system

$$\left(\partial_x^2 + \partial_y^2 + \partial_z^2\right)\mathcal{V}(x, y, z) = 0 \tag{2.7.10}$$

we further assume that the solution of $\mathcal{V}(x, y, z)$ takes to form of

$$\mathcal{V}(x, y, z) = X(x)Y(y)Z(z) \tag{2.7.11}$$

where $X(x), Y(y), Z(z)$ are depends only on x, y, z, respectively. For one-dimensional (1D) or two-dimensional (2D) problems, variables unrelated to the geometry can be omitted. For instance, in a 2D problem, the potential may be expressed as a separable function $\mathcal{V}(x, y) = X(x)Y(y)$. Substituting (2.7.11) into (2.7.10), and assuming $X(x)Y(y)Z(z) \neq 0$, we get

$$\frac{1}{X(x)}\frac{d^2X(x)}{dx^2} + \frac{1}{Y(y)}\frac{d^2Y(y)}{dy^2} + \frac{1}{Z(z)}\frac{d^2Z(z)}{dz^2} = 0 \tag{2.7.12}$$

Now each of the term in (2.7.12) depends on one single variable only; in order to let the equality hold, each term must be a constant value. Thus, let us set

$$\frac{1}{\Xi(\xi)}\frac{d^2\Xi(\xi)}{d\xi^2} = -k_\xi^2 \tag{2.7.13}$$

where $\Xi \coloneqq X, Y, Z$ and $\xi \coloneqq x, y, z$ with the constraint

$$k_x^2 + k_y^2 + k_z^2 = 0 \tag{2.7.14}$$

Then, a wise choice of the solution to (2.7.13) may take one of the following forms:

$$\Xi(\xi) = A\sin\left(k_\xi\xi\right) + B\cos\left(k_\xi\xi\right) \tag{2.7.15a}$$

$$\Xi(\xi) = Ae^{ik_\xi\xi} + Be^{-ik_\xi\xi} \tag{2.7.15b}$$

$$\Xi(\xi) = A\sinh\left(k_\xi\xi\right) + B\cosh\left(k_\xi\xi\right) \tag{2.7.15c}$$

depending on the boundary conditions given. Since the coefficients A and B may be complex, (2.7.15a-c) are convertible between each other. By applying the propriate boundary conditions, one can find the allowed values for k_ξ and the coefficients A and B. Let us illustrate this by the following example.

□ **Example 2-15** (Cheng 4-6)

Two grounded, semi-infinite, parallel-plane electrodes are separated by a distance d. A third electrode, oriented perpendicular to and electrically insulated from both, is held at a constant potential $\mathcal{V}_0$. Determine the potential distribution within the region bounded by these electrodes.

Solution

This is a 2D problem. Let

$$\mathcal{V}(x, y) = X(x)Y(y)$$

with the boundary conditions:

$$\begin{gathered} \mathcal{V}(0, y) = \mathcal{V}_0 \\ \mathcal{V}(\infty, y) = 0 \\ \mathcal{V}(x, 0) = 0 \\ \mathcal{V}(x, d) = 0 \end{gathered}$$

Solve $Y(y)$ first. By noting that we have vanishing results at $y = 0$ and $y = d$, we can write

$$Y(y) = C \sin(k_y y)$$

Then,

$$Y(0) = 0, \qquad Y(d) = C \sin(k_y d) = 0$$

which implies that

$$k_y = n\pi/d$$

and

$$Y_n(y) = C_n \sin\left(\frac{n\pi y}{d}\right), \qquad Y(y) = \sum_{n=0}^{\infty} Y_n(y)$$

where $n \in \mathbb{Z}^+$. Note that each n corresponds to a solution mode; thus, the constant values are also a function of n. We will express this explicitly in the following derivation. From (2.7.14) with $k_z = 0$, we have

$$k_x = \sqrt{-k_y^2} = in\pi/d$$

Thus

$$X_n(x) = A_n e^{ik_x x} + A_n e^{-ik_x x} = A_n e^{-n\pi x/d} + B_n e^{n\pi x/d}$$

Applying the boundary condition $\mathcal{V}(\infty, y) = 0$, it is required that $B = 0$, which leads to

$$\mathcal{V}(x,y) = \left(\sum_{n=0}^{\infty} C_n \sin\frac{n\pi y}{d}\right)\left(\sum_{n=0}^{\infty} A_n\, e^{-\frac{n\pi x}{d}}\right)$$
$$= \sum_{n=0}^{\infty} D_n\, e^{-\frac{n\pi x}{d}} \sin\left(\frac{n\pi y}{d}\right)$$

To find the coefficients D_n, let us apply the boundary condition $\mathcal{V}(0, y) = \mathcal{V}_0$ and the orthogonal property of sine functions

$$\sum_{n=0}^{\infty} \int_0^d D_n \sin^2\left(\frac{n\pi y}{d}\right) dy = \int_0^d D_n \sin^2\left(\frac{n\pi y}{d}\right) dy = \frac{d}{2} D_n$$
$$= \int_0^d \mathcal{V}_0 \sin\left(\frac{n\pi y}{d}\right) dy = \frac{\mathcal{V}_0 d}{n\pi}(1 - \cos n\pi)$$
$$= \begin{cases} \dfrac{2\mathcal{V}_0 d}{n\pi}, & n \text{ odd} \\ 0, & n \text{ even} \end{cases}$$

Thus,

$$D_n = \begin{cases} \dfrac{4\mathcal{V}_0}{n\pi}, & n \text{ odd} \\ 0, & n \text{ even} \end{cases}$$

2.8 Method of Images

The method of images is a technique used in electrostatics to simplify problems involving conductors and boundary conditions. It replaces the original configuration, which may include conductors, with an equivalent system of image charges placed in such a way that the boundary conditions are satisfied. The method is particularly useful for solving problems involving point charges near grounded conducting planes or spheres,

where direct application of Coulomb's law or related equations would be difficult due to complex boundary conditions.

The simplest case involves a point charge near an infinite grounded conducting plane. Suppose a point charge q is placed a distance d above the plane, which lies in the *xy*-plane and is held at zero potential. Without loss of generality, the point charge is assumed to be located on the z-axis. The resulting surface charge distribution on the conducting plane is clearly difficult to determine directly. By placing an image charge $-q$ at a distance d below the plane—i.e., symmetrically located at a total separation of $-d$ from the original charge—the boundary condition on the conducting plane is satisfied. As a result, the conductor can be replaced by the image charge, and the resulting potential for the region $z \geq 0$ is identical to that of the physical setup

$$\mathcal{V}(x, y, z) = \frac{q}{4\pi\epsilon_0}\left(\frac{1}{r^+} - \frac{1}{r^-}\right), \qquad z \geq 0 \tag{2.8.1}$$

with

$$r^+ = \sqrt{x^2 + y^2 + (z - d)^2} \tag{2.8.2a}$$

$$r^- = \sqrt{x^2 + y^2 + (z + d)^2} \tag{2.8.2b}$$

It is obvious that from (2.8.1) and (2.8.2) that the potential on the plane is $\mathcal{V}(x, y, 0) = 0$. From (2.8.1), the resulting electric field in the region $z \geq 0$ can be found by (2.1.19). Notice that the (2.8.1) only gives the result for $z \geq 0$, the region of interest.

□ **<u>Example 2-16</u>** <u>(Cheng 4-3)</u>

A positive point charge q is positioned at distances d_1 and d_2 from two grounded, mutually perpendicular conducting half-planes. Determine the force on q due to the charges induced on the conducting surfaces.

Solution

Suppose the point charge is located at $(d_1, d_2, 0)$. To satisfy the boundary condition that the potential is zero on both grounded planes, we need three image charges in addition to the real charge:

Real charge	$(d_1, d_2, 0)$	q
Image 1	$(-d_1, d_2, 0)$	$-q$
Image 2	$(d_1, -d_2, 0)$	$-q$
Image 3	$(-d_1, -d_2, 0)$	q

It is then clear that by placing these image charges, we can have zero potential on the boundaries when the conducting planes are replaced. Thus, the net force exerted on the real charge is

$$\vec{F} = \frac{q^2}{4\pi\epsilon_0}\left[\frac{-1}{(2d_1)^2}\hat{x} + \frac{-1}{(2d_2)^2}\hat{y} + \frac{1}{(2d_1)^2 + (2d_2)^2}\left(\frac{d_1}{\sqrt{d_1^2 + d_2^2}}\hat{x} + \frac{d_2}{\sqrt{d_1^2 + d_2^2}}\hat{y}\right)\right]$$